2030-35: SCALING UP

TEJAS MARK-2: A beefier Tejas with first flight eyed for 2026, induction by 2032-33. Planned fighters: 120-200 (6-10 squadrons)

MRFA DEAL: Plan for 114 twin-engine 4.5-gen Multi-Role Fighter Aircraft (MRFA) still in play; potential induction by 2030-32 if deal with foreign partners gets fast-tracked

AMCA MARK-1 Five prototypes in development for India's first fifth-gen stealth Advanced Medium Combat Aircraft. Induction by 2035—two squadrons (40 jets) to use GE F414 engines initially



2035-47: THE NEXT-GEN PUSH

AMCA MARK-2 Enhanced stealth jet with powerful 110-120 kN engines. IAF wants 160-200 jets (7-10 squadrons) by 2047. Total AMCA count could hit 240-300

TEDBF: Twin-Engine Deck-Based Fighter, a naval offshoot of AMCA. About 45-60 jets planned for carriers like INS *Vikrant*, with induction from 2035-2040, potentially doubling as the IAF assets

NEXT-GEN WILDCARD: By 2040, the IAF aims to leap into sixth-gen tech—optionally manned jets, swarm drones, hypersonics—either via AMCA upgrades or a new platform. No firm orders yet, but the intent's clear



manoeuvrability and even heavier weapon load. Rafale, the most advanced jet the IAF has, is a French 4.5-gen multirole fighter

The 21st century saw the emergence of 5th-gen fighters—such as the US's F-35 and China's J-20—which ushered in a new era, in which stealth and sensor fusion have rendered traditional dogfights nearly obsolete

Now, 6th-gen fighters like the F-47 and J-36 promise another leap forward, incorporating AI, drone wingmen and hypersonic weaponry

As the high-powered committee works toward a solution, a critical debate revolves around the AMCA's production partner—whether to entrust state-owned Hindustan Aeronautics Limited (HAL) or bring in private industry. HAL has proposed a consortium, in which it will hold 50 per cent of the stake and rope in four private firms with 12.5 per cent stake each to oversee specific tasks. But the government seems inclined toward an industry partnership approach. Sources in the ADA indicate that HAL may have to compete with other private firms separately, pointing to its poor track record when it comes to joint ventures. This signals a potential shift in India's defence manufacturing, with several top-tier private players vying for a greater role in military aviation projects.

However, scepticism persists over private sector involvement. A senior air marshal points to the massive investment required to establish a fighter jet manufacturing facility, including land acquisition, infrastructure, skilled workforce recruitment and training. "It's a humongous task with huge investments," he tells INDIA TODAY, emphasising the need for a viable business model before private firms can take the plunge. Key concerns include control over the ADA's R&D, cost implications and the division of responsibilities between HAL and private manufacturers. Another air marshal expresses doubts about the ability of both HAL and Indian private players to deliver, calling it a choice "between the devil and the deep blue sea". "None of our private sector stalwarts in aviation have any experience in the final assembly and delivery of small or midsize aircraft. Maybe we need handholding from a foreign player to attain expertise in this complex task," he adds.

HAL, already burdened with multiple projects, faces mounting challenges of its own. It is producing 83 Tejas Light Combat Aircraft (LCA) Mk-1A—with deliveries already delayed due to engine shortages—and has received clearance for an additional 97. Simultaneously, it is tasked with developing four prototypes of the Tejas Mk-2, expected to make its maiden flight by 2026. On March 25, GE Aerospace finally delivered the first of 99 F404-IN20 engines to HAL for the Mk-1A. To avoid similar setbacks with the AMCA, the Prime Minister's Office (PMO) is stepping in to directly monitor the jet's engine development. While the AMCA Mk-1 will initially take off with the GE F414 (98 kN) engines, a more powerful 110-120 kN engine, crucial for the enhanced AMCA Mk-2 fighter, is being developed by the DRDO's Gas Turbine Research Establishment (GTRE). Plans are afoot for foreign collaboration, with companies from France, the UK, the US and Russia showing interest in co-developing the engine.

But the question about who will manufacture the AMCA still looms. "At least, there is clarity that HAL will produce the Tejas Mk-1A and Mk-2 fighters. But no one knows yet as to who will make the AMCA," says Air Marshal S.B.P. Sinha (retd), a former IAF deputy chief. Still, he defends the AMCA's relevance, clarifying that it was never designed to compete with the F-47 or J-36 but to provide India with a "very capable fifth-gen fighter". But the unveiling of these sixth-gen fighters underscores that the window for the AMCA to remain relevant is rapidly shrinking, and India has to turbocharge its production. ■